

The background is a vibrant blue gradient. It features several white, semi-transparent gears of different sizes, some of which are interlocking. A network of white lines with small circular nodes at the intersections is spread across the background, resembling a digital or neural network. Two human hands are visible: one at the top left, with the index finger pointing towards the gears, and another at the bottom center, with the palm facing upwards. A large, semi-transparent grey rectangle is positioned in the upper right, serving as a backdrop for the title text.

CS 470 Project Two Conference Presentation: Cloud Development

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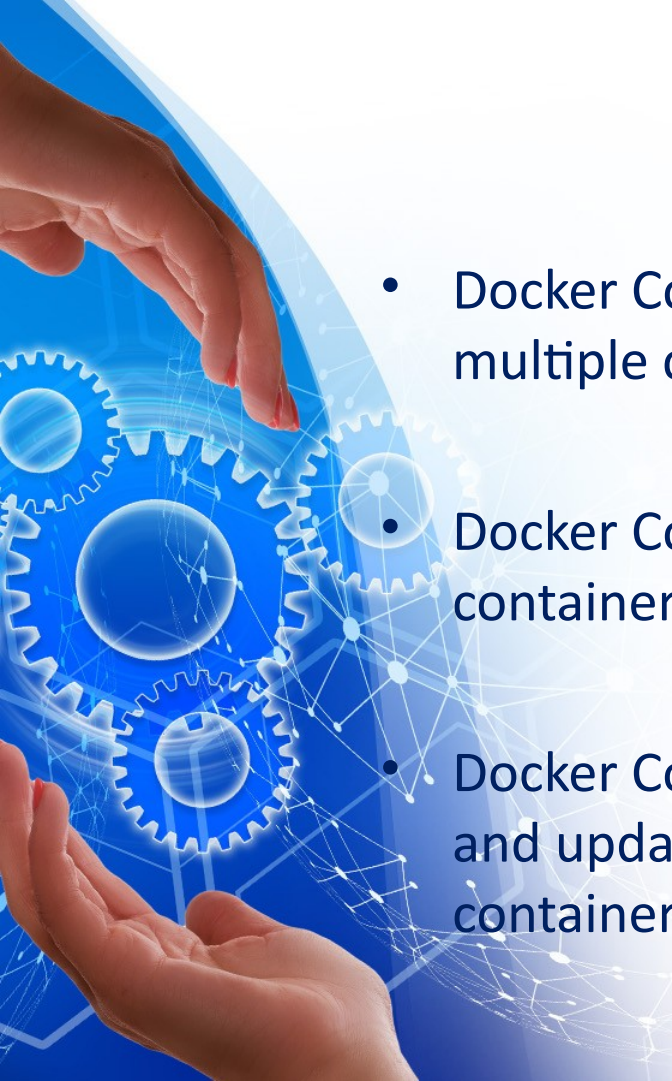
Overview

- I am Shawn Millin and am completing my Bachelor's Degree in Computer Science at Southern New Hampshire University .
- The purpose of this presentation is to articulate the intricacies of cloud development to both technical and nontechnical audiences.

A hand is shown interacting with a blue sphere. Inside the sphere, there are several interlocking gears of different sizes. A network diagram with nodes and connecting lines is also visible within the sphere. The background is a gradient of blue and white.

Containerization

- Containerization is the process of breaking a full-stack application into several smaller pieces which get placed into containers for deployment. Includes front-end and back-end processes
- A container platform such as Docker is a necessary tool used to make containers using container images. These contain all of the dependencies necessary for the program to run.
- Necessary tools also included MongoDB as a database and the use of Angular, ExpressJS, and Node.js (MEAN)



Orchestration

- Docker Compose allows a user to define and run multiple containers with a single command.
- Docker Compose uses a YAML file for the definition of a containerized component of an application.
- Docker Compose makes applications easier to manage and update, as well as, maintain scaling and launching of containerized services.



The Serverless Cloud

Serverless

- Serverless is a cloud computing model. The serverless cloud allows developers to focus on coding without worrying about managing the servers such as scaling and maintenance.
- AWS Amazon Web Services is a pay-as-needed cloud service provided by Amazon.
- S3 is service on AWS that uses cloud storage to store data in “buckets” that are accessible anywhere, to save and retrieve data, with internet connection and scale to almost unlimited size. In contrast, local storage on a physical device is limited to hardware and local or remote access and has a higher hardware investment.



The Serverless Cloud

API & Lambda

- The advantages of using a serverless API include the pay-as-you-go model instead of paying for a large chunk of data storage. Developers do not need to manage and maintain servers and security resulting in faster application releases. Serverless APIs automatically scale up or down.
- Lambda API Logic is an AWS service where developers submit code to Lambda functions which perform a specific tasks in response to events. Code can be written from the languages of Python, C#, Node.js, and Java.
- To integrate the frontend with the backend in AWS the Lambda functions need to be linked to API Gateways with Cross-Origin Resource Sharing (CORS)



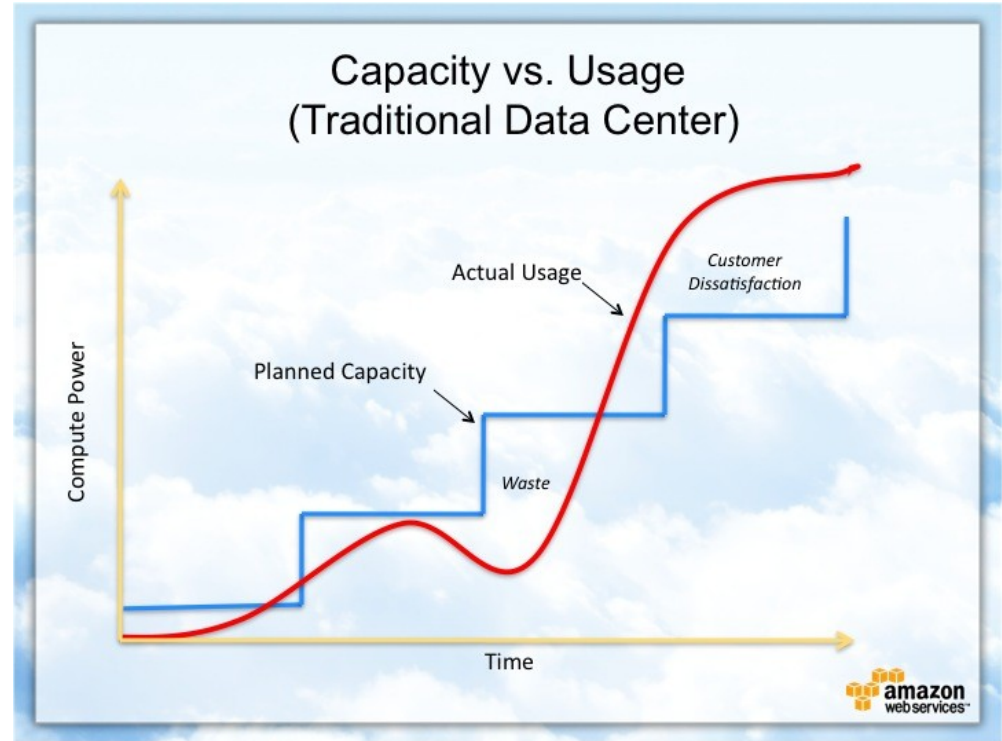
The Serverless Cloud

Database

- MongoDB and DynamoDB are NoSQL databases that both utilize the document-base but DynamoDB's model is key-value pairs with nested JSON-like documents while MongoDB is hierarchical data structured within a single document . MongoDB has a flexible schema while DynamoDB requires a predefined schema and designing tables with size limits.
- For this project queries were performed for finding all questions in the database table, finding a single question in the database, finding the answer in the database, and adding, updating, and deleting entries to the database.

Cloud-Based Development Principles

- Elasticity: The ability to scale up or down
- Pay-for-use model: Only pay for the amount of services needed





Securing Your Cloud App

Access

- Access is controlled through the Identity and Access Management (IAM) service for creation of roles and permissions

Policies

- A role is defined with policies. Each having different authorization levels. Roles are assigned to users.
- In this project a policy was coded to make the web page public

API Security

- The secure connection between Lambda and Gateway is API keys
- Between Lambda and the database is roles and policies
- S3 Bucket is secured by AWS



CONCLUSION

- Cloud development can scale easily when demand is increased or decreased.
- Cloud development through AWS can lower the amount of work by allowing the service to take care of maintenance and primary security
- Cloud development can lower the cost of application development with the pay-as-needed model.

Thank you for your time.